Chapter 7 - Loops in Python.

Sometimes we want to repeat a set of statements in our ~~python~~ program. for instance: Print 1 to 1000

Loops make it easy for a programmer to tell the computer, which set of instructions to repeat and how!

Types of loops in Python.

-> Primarily there are two loops in Python
1. While loop.
2. For loop.

Now we will look into these two one by one

While loop:

The syntax of a while loop looks like this:

```
While Condition:
    # Body of the loop
```

=> this block keeps executing until condition is true.

In while loops, the condition is checked first. If it evaluates to true, The body of the loop is executed, otherwise not! If loop is entered, the process of [Condition check & execution] is continued until the condition becomes false.

Quick Quiz: Write a program to print 1 to 50 using while loop.

Sol:
```
i = 0
while i < 51 :  => prints 1 to 50.
    Print (i)
    i += 1 or i = i + 1
```
This loop will print numbers from 1 to 50.

An Example:

```
i = 0
while i < 5        => Prints "micro" - 5 times
    Print ("micro")
    i = i + 1
```

Note: If the condition never become false, the loop keeps getting executed.

Quick Quiz: Write a program to print the content of a list using while loop.

Sol:
```
l = ["micro", "adil", False, 34, "abid"]

i = 0
while i < length (l) :
    Print (l [i])
    i += 1
```

For loop:
A for loop is used to iterate through a sequence like list, tuple or string (iterables)

The syntax of a for loop looks like this:

```
l = [1, 7, 8]
for item in l:
    print (item) → print 1, 7, & 8.
```

Range function in Python
The range function in python is used to generate a sequence of numbers.
We can also specify the start, stop and step-size as follows:

range (start, stop, step_size)

step_size is usually not used with range ()

An example demonstrating range () function

```
for i in range (0, 7): → range (7) can also be used
    print (i)          → print 0 to 6
```

For loop with else:
An optional else can be used with a for loop if the code is to be executed when the loop exhausts

Example :

```
l = [1, 7, 8]
for item in l:
    print (item)
else
    print ("Done") => This is printed when the loop exhausts !
```

Output :
1
7
8
Done

The Break Statement :
'break' is used to come out of the loop when encountered. It instructs the program to Exit the loop now.

Example :

```
for i in range (0, 80):
    print (i)        -> This will print
    if i == 3:          0, 1, 2 and 3.
        break
```

The Continue statement :
'Continue' is used to stop the current iteration of the loop and continue with the next one It instructs the program to "skip this iteration"

Example :

```
for i in range (4)
    print ("printing")
```

```
if i == 2:
    Continue
print (i)
```

⇒ if i is 2, the iteration is skipped.

pass statement :

pass statement is null statement in python
It instructs to "do nothing"

Example :

```
l = [1, 7, 8]
for item in l :
    pass
```

→ Without pass the program will throw an error

I-b Q. Write a program to print the star pattern

```
  *
 ***
*****
```

for n = 3

Sol

```
n = int (input ("enter your number"))

for i in range (1, n+1) :
    print (" " * (n-i), end = "")
    print ("*" * (2 * i - 1), end="")
    print (" ")
```

n-i : This ~~creates~~ calculates the number of spaces before the star (*) in each row. As i increases, the number of spaces decreases.
- for example, if n=5 and i=1, then n-i=4, so 4 spaces are ~~need~~ printed
- if i=2 then n-i=3, so 3 spaces are ~~need~~ printed.

2 * i - 1 : this calculates the number of stars (*) to print in each row, the number of stars increases as i increases.
- for example, if i=1, then 2*1 - 1 = 1, so 1 star is printed.
- if i = 2 then, 2*2 - 1 = 3, so 3 stars are printed.